

ECE240:PCS**Quiz (Comp)***Instructor: Ranjitha Prasad*

Please provide your answers clearly, with all the steps outlined. Total: 15 marks.

Use $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$, and $\text{sinc}(\pi t) \leftrightarrow \Pi(f)$, where $\Pi(\cdot)$ is as defined in class (rectangular function between -0.5 to 0.5, with unit magnitude).

Problem 1 Analog Modulation : Answer the following:

1. A double side-band amplitude modulated signal is modulated by the message signal given by $m(t) = 2\cos 2000\pi t + \cos 6000\pi t$. The modulated signal is given by $u(t) = 100m(t)\cos(2\pi f_c t)$, where $f_c = 1\text{MHz}$. Carefully sketch the frequency spectrum and indicate all frequency components. Write the block diagram for the demodulator (mention the one that best fits such a modulated signal) and carefully mark all components in the block diagram (There is no need to write the equations).(2)
2. The message signal $m(t) = 10\text{sinc}(400\pi t)$ frequency modulates a carrier $c(t) = 100\cos(2\pi f_c t)$. The modulation index is 6. Write the expression for the modulated signal $u(t)$, and find the power and bandwidth of the signal. What is the maximum frequency deviation of the modulated signal? ... (2)

Problem 2 A signal $g(t) = \text{sinc}(5\pi t)$ is sampled (using uniformly spaced impulses) at a rate of: (i) 5 Hz; (ii) 10 Hz. For each of the cases sketch the spectrum of the sampled signal..(2)

Problem 3 A random process is defined by $Y(t) = T + (1 - t)$ where T is a uniform random variable in $(0, 1)$. Answer the following:

1. Find the mean $m_Y(t)$, the autocorrelation $R_Y(t_1, t_2)$(2)
2. Find the cumulative distribution function of $Y(t)$... (2)

Problem 5 Consider the di-code line-code. In this line-code, for each transition of the input there is an output pulse (covering entire T_b duration) of opposite polarity from the preceding pulse. For no transition between input pulses the output is 0. Hence, the possible levels in this linecode are 1, 0 and -1. Encode the sequence 110101111001. (1)

Problem 6 A baseband PAM system uses a pulse shape $p(t)$ given by:

$$p(t) = \begin{cases} 1 - \frac{|t|}{T} & |t| \leq T \\ 0 & |t| > T \end{cases}$$

Does $p(t)$ fulfil the Nyquist criterion? Present the eye diagram at the output of the filter...(2.5)

Problem 7 Consider a coin tossing experiment. If the probability of occurrence of tail is $p = 0.5$, what is the entropy of the resulting event. Repeat the calculation for $p = 0.7$. Which among the two is a case of higher uncertainty? Does higher uncertainty correspond to higher information? Explain your answer. ... (2.5)